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The integration of explainable AI in Educational Data Mining for student academic performance prediction and support system

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ABSTRACT

The academic performance of students is important because it is a crucial indicator of their educational life. The absence of prior experience, guidelines, and social and personal factors can contribute to unforeseen issues that adversely impact a student's academic performance. An Artificial Intelligence (AI)-driven Educational Data Mining (EDM) system can serve as a solution in this regard. In this study, we proposed an EDM system that integrates machine learning to classify student academic performance and explainable AI techniques for explainability. We utilize standard ML models such as Decision Tree and Random Forest, and Boosting models such as Gradient Boosting and Extreme Gradient Boosting (XGB) for multiclass classification, and SHAP, Shapash, Eli5, and LIME as XAI techniques for both global and local explainability. For the student's academic performance classification task, the XGB model outperforms other models and achieved 83% accuracy. The analysis indicates that the evaluations of the curricula, courses, and economic variables of different semesters greatly impact the student's academic performances. To provide proper direction to students, we also developed a support system that uses the trained model from our analysis and provides recommendations and suggestions based on the input data. The further study integrates real-time data and mobile applications for the support system.

1. Introduction

Education is the basis of the development of a country. However, educational institutions still have the most common problem of students dropping out in unexpected ways. Perhaps this problem is because no teaching and learning system perfectly suits all categories of students. In addition, the way of teaching and the medium of class play a role in causing this problem [1].

There are a few major concerns about the prospect of educational institutes such as if the student is about to pass or not, what kind of marks they are about to obtain, and because of that are going to drop out or not [2,3]. If there is a system that could predict not only dropout rates but also the difficulties students face in obtaining good scores, it could benefit not only students but also institutions [4–6]. The EDM (Educational Data Mining) techniques are capable of performing these tasks [7,8].

Machine Learning(ML) derives rules from training examples to develop classifiers which enable the categorization of unseen instances [9]. The practical implementation of ML in education has become increasingly popular recently [10]. This is mainly because it can provide valuable information and predictions related to student academic performance [11–13]. Predicting and analyzing students' performance is essential for educators to identify weaknesses and help students improve their scores. It also encourages students to improve their learning activities [11]. ML techniques can predict student performance, and students at risk can be identified as early as possible so that appropriate steps can be taken to improve their performance [14].

The academic performance of students is influenced by several variables, including previous semester results, previous academic performance, financial situation, family background, secondary school grade, study and travel time, department, family size, courses, annual income, parent's educational background and occupations, midterm exam performance, etc. [9,13–15]. These factors can influence academic performance either negatively or positively. Hence, there has been a lot of research in the field of EDM to uncover external factors that impact students' academic success or failure [16]. Sometimes,

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the impact of negative factors is so awful that a student can even be dropped out during the study period, either in high school or in university. It can be a cause of termination of the academic career of a student due to lack of fulfillment of the above-mentioned properties. In addition to this, universities and institutions are trying to improve the academic performance of their students by finding strategies to reduce or eliminate the impact of negative factors on their students [16]. Existing works focus on classifying the performances and prediction of students' performances, but still, a gap in EDM to find the cause behind these performances can be a good research direction. In this study, we try to find the cause behind this problem by considering the following objectives.

- We proposed an ML-enabled EDM approach to anticipate the academic performance of students at an early stage.
- We find out the major factors behind a student's academic performance using XAI techniques that show both the local and global stories of black-box ML models in our proposed setup.
- We also proposed a decision support system that identifies the categories of the students based on real-time data and provides personalized support according to their current state.

The rest of the paper is organized as in Section 2 the related works, in Section 3 the proposed methodology with details of tools and techniques, in Section 4 the results and discussion, and finally in Section 6 the conclusion and future research direction.

2. Literature review

Anal Acharya and Devadatta Sinha developed five classes of machine learning algorithms (MLAs) to predict student performance at an earlier stage. 309 and 104 data samples were collected for training and testing purposes, respectively, from a group of students majoring in computer science from some undergraduate colleges in Kolkata. 14 independent attributes were used, including past academic history, financial state, the status of a family, mid-semester test performance, etc. C4.5, MLP, NB, 1NN, and SMO algorithms were used and got the best result of 79% for training and 66% for testing. This proposed system can be improved by considering more factors and increasing efficiency by approaching Combined Multiple Classifiers (CMC) [9].

Mojisola G. Asogbon et al. [13] developed a multiclass support vector machine (MSVM) predictor to predict student performance at different levels at the University of Lagos, Nigeria. Among 300 data sets, 250 samples were used for training purposes and 50 for testing. The proposed algorithm achieved the non-cross-validation MSVM had an average testing accuracy of 66%, while the cross-validation MSVM outperformed it with a score of 72.25%. This algorithm predicts the 4 classes of students out of 5 classes. This is the drawback of this algorithm. Hussein Altabrawee et al. [14] predicted the performance of computer science students offered by Al-Muthanna University (MU), College of Humanities by using five algorithms named Artificial Neural Network (ANN), Naïve Bayes (NB), Decision Tree (DT), and Logistic Regression (LR). The dataset collected around 2015 to 2016 contains a total of 161 students where 76 are male and 85 are female students' data. Among these five algorithms, the best model is ANN whose have a 0.807 ROC index and 77.04% accuracy.

The authors of [16] analyzed the performance of the students in an online course using Neutral Networks. The data set was collected from 7909 students who had participated in a series of four online courses. The datasets obtained 39 columns and 7009 rows and got 82.46% of accuracy. The 'Random Forest' technique was also applied to identify the variables with the highest weight. This model can be used to improve the academic performance of the student. Likewise, the authors of [17] introduced a hybrid ML model based on four well-known algorithms named support vector machine using the radial basis function kernel (SVMRBF), naïve Bayes (NB), decision tree C5.0, and random forest (RF). The datasets were named GDS1 (2000 samples), GDS2 (4000 samples), and ADS3 from 22 high schools in Cambodia consisting of 1204 samples. They observed the student's performance by three methods and compared each other. At first, the performance of those baseline methods was observed. Secondly, performance was advanced with 10-CV. Lastly, they used a combination of the PCA to baseline models and the 10-CV method to improve the classification performance. After comparing all these methods based on classification accuracy and RMSE as measurement parameters, it is found that the proposed hybrid model was the most satisfying method.

Monica et al. [20] employed ML approaches to identify students who are at risk of failing in an early phase of their academic path so that support mechanisms can be implemented. They help lower academic failure rates in higher education. They discovered that boosting algorithms outperform conventional approaches for the particular categorization work. Radhika R Halde [25] approaches with the wise use of data analytics and different machine learning algorithms on the academic performance rate of a student. Their analysis in academics had two applications, retention and prediction of student performance. The neural networks and support vector machines were more precise for numerical predictions. The results of this project were not up to their expectations. An accuracy of 60%–80% was recorded. The accuracy needs to be increased, which is possible by knowing the personal learning styles of the students. An article [1] uses supervised ML and data mining to detect the dropout rate and academic performance. Their proposed model is based on classification to make predictions about the future performance of the students. To build the model, a data mining approach is used in the classification technique, and algorithms such as the support vector machine and logistic regression are applied to their project. The accuracy of their project is around 80%. The results show that performance depends on the medium of the classes and the way of teaching.

Students dropping out before exams creates a loophole for the future progress of a nation. Early prediction of tentative dropout can alleviate the problem. The EDM is a useful tool for educational institutions. It is a common research area for exploring the fields of education using ML and data mining techniques and approaches. Weka a common ML application is used in the project which also supports different types of DM tasks. The exact results were achieved using logistic regression with accuracies of 68.7% and 88.8% [19]. Iti Burman et al. [18] built a model using data mining together with a support vector machine, neural network, and decision tree to predict the annual performance of students. The decision tree predicts 65% accuracy, whereas the SVM predicts with an accuracy of 83%. The SVM supports the model better than others. It is a supervised learning technique that focuses primarily on classifying data. The dataset for their project consists of 1000 student records. They used 70% of the data to train their model and used the remaining 30% for testing. The paper focuses on non-intellectual parameters to predict the outcome. The result of their project is much more accurate, which is approximately 90%. The authors of [21] aim to predict students at high risk. They used classification algorithms in their subsequent work. In the project, the dataset was taken from the student information system of the State University of Turkey. Several machine learning algorithms are used to analyze the results. After analyzing, the proposed model shows an accuracy of 70%–75%.

3. Methodology

3.1. Approach overview

Our main objective is to develop an XAI-integrated EDM system that can analyze the student's academic performance and find the reason behind it. At the beginning of the data preprocessing phase 1, we detect the outlier and remove it, then the object data field encodes into numeric values to convert the dataset into more ML-centric. An EDA (Exploratory Data Analysis) system is integrated with the Chi-square

Table 1
Related existing work comparison.

Ref.	Year	Dataset	Algorithms applied	Metrics used	Best algorithm	XAI
[9]	2014	Few Undergraduate Colleges in Kolkata, data has 15 attributes and 413 samples	C4.5, MLP, NB, 1NN, SMO	F-measure, Kappa Statistic	C4.5 (F-measure 66%)	Not applied
[13]	2016	University of Lagos, Nigeria. Data having 17 attributes and 300 samples	Cross-Validation MSVM, Non-cross Validation MSVM	Accuracy, Error Rate	Cross-Validation MSVM (Accuracy 72.25%)	Not applied
[14]	2019	The College of Humanities at Al-Muthanna University, dataset having 20 attributes and 161 samples	ANN, DT, LR, NB	Precision, Recall, F-Measures, Accuracy, Classification Error	ANN (Accuracy 77.04%)	Not used
[16]	2019	Data collected online course survey, data contains 39 attributes and 7909 samples	RF, LR, KNN, NN	Precision, Recall, F1-Score, MSE	NN (Accuracy 82.46%)	Model's features weight
[18]	2019	1000 Records based on 29 non intellectual constructs of students	KNN, NB, SMV (Linear Kernel), SMV (Radial Basis Function Kernel)	Accuracy, Kappa	SMV(Radial Basis Function Kernel) (Accuracy 90%)	Not applied
[17]	2020	22 High Schools in Cambodia, dataset having 43 attributes and 7204 samples	RF, C5.0, NB, SVMRBF, Hybrid RF	Classification Accuracy, RSME	Hybrid RF (Accuracy 99.72%)	Not applied
[1]	2020	Data collected at an e-Learning System and it has 16 attributes and 500 samples	LR, SVM	Recall, Precision, F1-score, Accuracy	SVM (Accuracy 79.3%)	Not applied
[19]	2020	The College of CS and IT, University of Basra. Data has 8 attributes and 499 samples	DT, NB, LR, SVM, KNN, SMO, NN	Precision, Recall, AUC, ROC	LR (Accuracy 88.8%)	Not applied
[20]	2021	Polytechnic Institute of Portalegre, Portugal. Data have 36 attributes and 4424 samples	LR, SVM, DT, RF, XGB	F1-score, Accuracy	XGB (Accuracy 73%)	Not applied
[21]	2022	1854 students who took the Turkish Language-I course in a state University in Turkey	RF, NN, LR, SVM, NB, and KNN	Accuracy, Precision, AUC	RF and NN (Accuracy 74.6%)	Not applied
[22]	2023	Data collected at Kaggle website, it has 15 attributes and 215 samples	LR, KNN, SVM, NB, DT, Ensemble	Accuracy Recall, F1-score	NB (Recall 91.2%)	Few applied
[23]	2024	Same as of [20]	Adaboost, RF, DT	Accuracy, Precision, AUC, Recall	Adaboost (Accuracy 77%)	SHAP
[24]	2024	Data collected at BME, Hungary. It has 8508 samples	LR, XGBoost, Catboost	Precision, AUC	Catboost (Avg. Precision 84.7%)	SHAP, LIME
Ours	2024	Same as of [20]	DT, RF, GB, and XGB	Confusion Matrix, Accuracy, Precision, Recall, F1-score	XGB (Accuracy 83%)	SHAP, LIME, Eli5

test in our methodology that finds the association of the important variables with the target variable shows an important insight and enhances the understandability of the dataset. In preprocessing phase 2, we select the top 20 potential features using the Extra Tree Classifier (ETC) and use these features for model training. These 20 features are 'Curricular units 2nd sem (approved)', 'Curricular units 2nd sem (grade)', 'Curricular units 1st sem (approved)', 'Tuition fees status', 'Course', 'Curricular units 1st sem (grade)', 'Scholarship status', 'Age at enrollment', 'Curricular units 2nd sem (evaluations)', 'Curricular units 1st sem (evaluations)', 'Father occupation', 'Mother qualification', 'Rate of Unemployment', 'Mother occupation', 'GDP', 'Father qualification', 'Inflation rate', 'Curricular units 2nd sem (evaluations)', 'Curricular units 1st sem (evaluations)', 'Application mode', 'Admission grade'. For the development of the ML model, we divide the preprocessed dataset into training (80%) and testing (20%) portions as the general statement of the development of the ML model. To reduce the bias of the ML classifiers, we handle the imbalance properties of the training data using SMOTE. After that, data normalization is performed. We then apply the Standard Model (DT and RF) and the Boosting Model (GB, XGB) to develop an EDM system. We evaluated the performance of the individual algorithm using Accuracy, Precision, Recall, and F1 Score to find the best-performing model for EDM and explainability. Finally, we apply some XAI tools to find the global and local explainability of the ML model which is like a black box, and explain the insights of the dataset concerning the best-performing model. The overview of the proposed EDM system is in Fig. 1. To find the stability of the EDM system, we also find the result of the (70:30) and (50:50) ratio of training and testing of the dataset in the significant test section. A brief comparison of the proposed work with the related existing works is given in Table 1.

3.2. Dataset description

The student academic performance dataset is collected from online [20]. It contains 4424 instances, 36 independent variables, and a target variable. The target variable has three classes (Dropout, Relative Success, and Absolute Success) that indicate the academic performance status of the students. In other words, the academic performance status

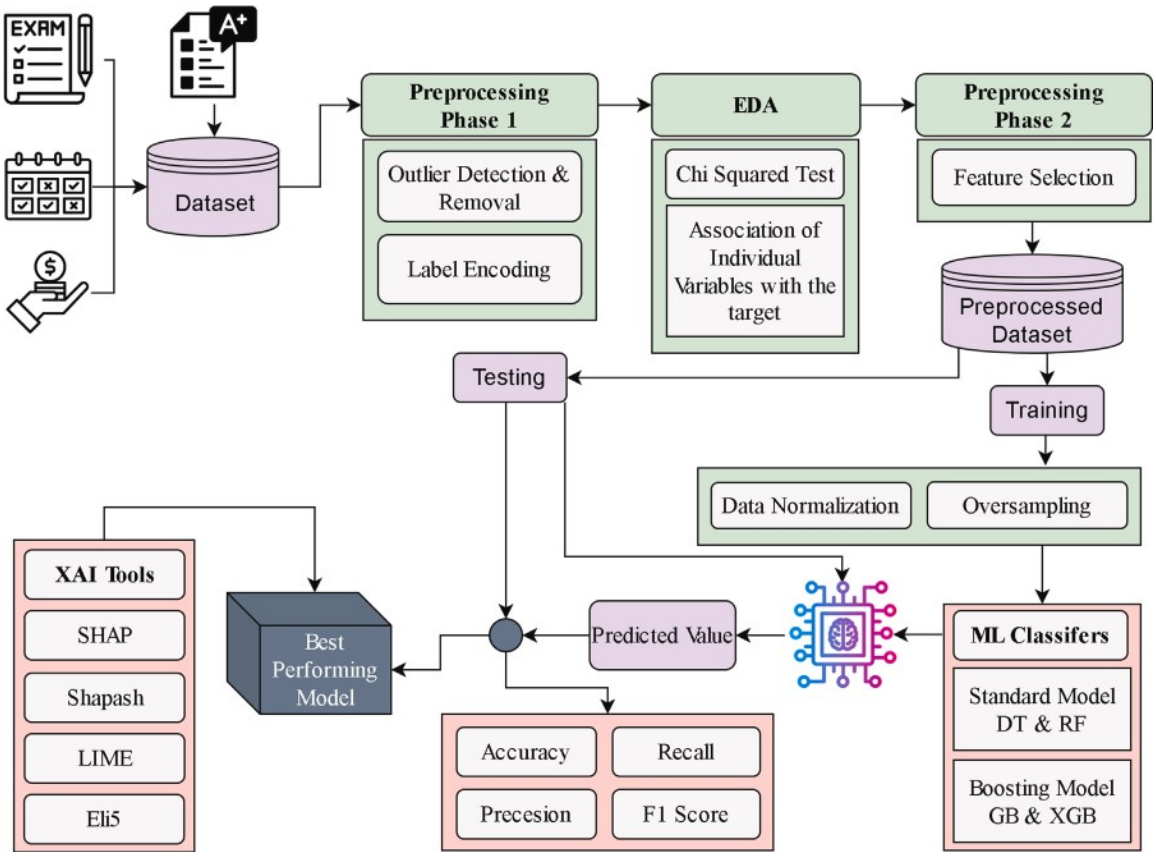


Fig. 1. Workflow of the proposed methodology with data preparation, model development, and explainability.

of the students represents a high-risk, medium-risk, and low-risk student accordingly. In addition, the dataset contains student information during enrollment, such as academic path, demographics, social factors, and information about their academic performance in the first and second semesters. The details of the characteristics are in Table 2.

3.3. Preprocessing

In this pipeline, first, we have to check whether the target variable is balanced or imbalanced. The results show that it is an imbalanced problem. Since the dataset is not large, the oversampling technique SMOTE is used to balance data instead of the undersampling balancing technique, where there is a chance of loss of information. After that, we conduct the data analysis part. In this part, we perform a chi-square test between some categorical variables and the target variable. The plot helps to understand the underlying association between an independent variable and a target variable. The result of the chi-square test is shown in the bar graph. Following that, we check if any missing values exist. It returns that there is no missing value in the data. After that, we check for outliers and identify them using the following formula [26] and replace the lower outlier with the value of the lower limit and the upper outlier with the upper limit value.

$$IQR = q_3 - q_1$$
$$LL = q_1 - (1.5 \times IQR)$$
$$UL = q_3 + (1.5 \times IQR)$$

where, IQR is interquartile value, q_1 is first quartile, q_2 is second quartile, q_3 is third quartile, LL is lower limit and UL is upper limit.

Consequently, we convert the non-numeric value to a numeric value. Afterward, select the top 20 features for the model building. Following that, we scale the data with the normalization scaling technique, which decreases the range of the data and helps the models capture the pattern precisely. It obtains the dataset has no missing values.

SMOTE: One method of addressing unbalanced data in the field of machine learning is called SMOTE. It equalizes imbalanced data by creating synthetic samples for the minority category [27]. It transforms the initial dataset in a way so that the resulting dataset gains more information while not removing any information from the original dataset, allowing the predictive models to acquire knowledge with equality for each category and reducing discrimination in predicted outcomes.

3.4. Experimental setup

Table 3 explains the experimental setup of our proposed method. In the Exploratory Data Analysis (EDA) phase, we employed IBM SPSS Statistics to perform analyses of the dataset. Specifically, we used the Chi-square test to identify correlations and associations among the variables. The subsequent task, referred to as ML-based Analysis, is executed using Python 3.8, along with Pandas, Matplotlib, and Scikit-learn. This stage encompasses handling missing values, detecting and removing outliers, encoding, data balancing, feature selection, data scaling, model development, and evaluation. To conclude, the "Explainability" task is addressed using SHAP, Shapash, Eli5, and LIME. This stage involves conducting Feature Importance analysis, assessing Features Weight and tolerance, evaluating Permutation Feature Importance, generating Summary Plot, Contribution Plot, Instance Explanation, and Comparison Plot.

3.5. Methods used in the study

3.5.1. Decision Tree (DT)

DT is an ML model used for classification and regression problems. It splits the data based on conditions at each node, leading to a final decision at the leaf nodes, which determines the model's prediction [28].

Table 2
Description of the variables used in this analysis.

Variable Name	Description
Gender	Male or Female
Age at enrollment	Age of student at enrollment
Admission grade	Admission grade in the range between 0 and 200
Scholarship status	Whether the student secured a scholarship or not
Displacement status	whether the student displaced or not
Debt status	whether the student debtor or not
Tuition fees status	Tuition fees of the student paid or not
Previous qualification	Previous academic degree achieved by the student
Previous qualification (grade)	Grade achieved by the student at last completed degree
Nationality	Nationality of the student
Curricular units 1st sem (credited)	Number of curricular units credited in the 1st semester
Curricular units 1st sem (enrolled)	Number of curricular units enrolled in the 1st semester
Curricular units 1st sem (approved)	Number of curricular units approved in the 1st semester
Curricular units 1st sem (grade)	Grade average in the 1st semester (between 0 and 20)
Curricular units 1st sem (evaluations)	Number of evaluations to curricular units in the 1st semester
Curricular units 1st sem (without evaluations)	Number of curricular units without evaluations in the 1st semester
Curricular units 2nd sem (credited)	Number of curricular units credited in the 2nd semester
Curricular units 2nd sem (approved)	Number of curricular units approved in the 2nd semester
Curricular units 2nd sem (evaluations)	Number of evaluations to curricular units in the 2nd semester
GDP	Gross Domestic Product
Rate of unemployment	Rate of unemployment (%)
Inflation rate	Inflation rate (%)
Application order	Application order (between 0 - first choice; and 9 last choice)
Application mode	At which phase the student applied
Father qualification	Highest Educational degree achieved by the father of the student
Mather qualification	Highest Educational degree achieved by the mother of the student
Father occupation	Father of the student work at which profession
Mother occupation	Mather of the student work at which profession
Marital Status	The student is married or single or divorced or separated
Course	The course is taken by the student
Daytime/evening attendance	Either the student presented in class in daytime or evening
Target	Three category classification task

Table 3
Experimental setup.

Tasks Performed	Task Description	Tools and Techniques
Exploratory Data Analysis	Correlation and association among variables using Chi-squared test	IBM SPSS Statistics
ML-based Analysis	Missing value handling, Outlier detection and removal, Encoding, Data balancing, Feature Selection, Data Scaling, Model development and evaluation	Python 3.8, Pandas, Matplotlib, Scikit-learn
XAI	Feature Importance, Features Weight and tolerance, Permutation Feature Importance, Summary Plot, Contribution Plot, Instance Explanation, Comparison Plot	SHAP, Shapash, Eli5, LIME

Each sample follows a path from the root node to a single leaf node, which contains the prediction for that sample. The prediction for an input $x(i)$ can be expressed as

$$Y(i) = V_{\ell(i)} \tag{1}$$

Where, $V_{\ell(i)}$ represents the value assigned to the leaf node $\ell(i)$ that the input $x(i)$ falls into. $\ell(i)$ represents the index of the leaf node corresponding to $x(i)$.

The DT is quick to train models, does not need scale, and is simple to understand. Model overfitting occurs when DT acquires more information, such as noise, which impacts how well the model generalizes [29].

3.5.2. Random Forest (RF)

RF is a popular ensemble-supervised ML approach that is applied to both classification as well as regression. It addresses the issues with generalizing models that arise with a single DT [30]. It integrates various DTs, in which each tree is constructed differently based on the selection of a random sample, gathers the results of each model, and then categorizes data points according to the models' majority of votes [31]. In almost all cases, RF outperforms an individual DT. It operates badly when dealing with a large-dimensional dataset that is sparse. The final prediction of RF can be represented mathematically as:

$$Y(i) = \frac{1}{N}\sum_{1}^{N} DT_N(X(i)) \tag{2}$$

Where $Y(i)$ represents the prediction for the input $X(i)$ and $DT_N\,X(i)$ represents the prediction of the Nth Decision Tree.

3.5.3. Gradient Boosting (GB)

One of the most effective supervised classification algorithms is GB. It is an ensemble-boosting method. GB operates by reducing the tendency to make mistakes [32]. It can work well with multidimensional and complex datasets because of its predictive power and speed. GB makes use of the base estimator's predetermined numbers. The mathematical expression of GB is

$$Y_i^{(t)} = Y_i^{(t-1)} + RF_t(x_i) \tag{3}$$

Where $Y_i^{(t)}$ represents the prediction at iteration t, R represents the learning rate that controls the contribution of each weak learner, and $F_t(x_i)$ represents the new weak learner's prediction which is added at iteration t.

3.5.4. Extreme Gradient Boosting (XGB)

The ensemble-supervised approach XGB employs boosting methods. Both classification and regression tasks can be accomplished with XGB [33]. In XGB, trees are linked sequentially, where each new tree is added at each iteration to correct the mistakes made by the previous ensemble. A key aspect of XGB is its regularization techniques, which help prevent overfitting, distinguishing it from simpler GB methods. Additionally, the learning rate plays a crucial role in controlling the model's sensitivity to parameter settings during training [34]. The general mathematical representation of XGB is

$$Y_i^{(t)} = Y_i^{(t-1)} + F_t(x_i) \tag{4}$$

Where $Y_i^{(t)}$ is the prediction at iteration t, and $F_t(x_i)$ represents the new tree's prediction added at iteration t. Unlike standard gradient boosting, XGBoost also incorporates techniques such as L1/L2 regularization and handling of sparse data, contributing to its robustness and efficiency in various tasks.

3.6. Performance metrics

Accuracy: Accuracy is a fundamental performance statistic for ML models. It indicates how accurate the trained model's predictions are. Accuracy is estimated by dividing the total sample count by the number of correct predictions. The percentage of accurately predicted instances to all instances is known as accuracy.

$$Accuracy = \frac{Correct\ predictions}{Total\ predictions} \tag{5}$$

Precision: One class-wise performance measurement metric used in ML is precision, which gives a clear picture of how well the predictive models being built are performing in predicting each class of the problem. The proportion of true positives (TP) to all of the positive predictions the model made is known as precision.

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

Precision is a useful performance measuring parameter when correct false positive predictions are more essential than false negatives in ML tasks.

Recall: Another class-wise performance measurement metric used to assess ML models is recall. It is also referred to as the actual positive rate or the sensitivity of the model. The term recall is measured by the division of the total number of actual positive cases in the data with the number of true positives (TP). When more precise false negative predictions are valued over false positives (FP), it is a useful metric.

$$Recall = \frac{TP}{TP + FN} \tag{7}$$

F1 Score: A composite measurement statistic called the F1-score combines the metrics of recall and precision. It is determined by averaging precision and recall harmonically.

$$F1\ Score = \frac{2 * (Precision * Recall)}{Precision + Recall} \tag{8}$$

It is beneficial when anticipating outcomes accurately is crucial for both positive and negative classes in the classification task. In a problem of imbalanced classification, it may serve as a more useful metric than accuracy.

3.7. XAI techniques

SHAP: SHAP (Shapley Additive Explanations) is a popular explainability method grounded in cooperative game theory. It computes Shapley values, which attribute the contribution of each feature to the final prediction of the model. By doing so, SHAP offers insight into the importance and role of individual features, fostering both transparency and trust in the model's decision-making process. It is widely used in ML applications for interpreting complex models and ensuring accountability [35].

SHAPash: SHAPash builds upon SHAP by offering enhanced interpretability through an interactive interface and more comprehensive reporting. While SHAP provides detailed explanations of model behavior on an instance-by-instance basis, SHAPash improves on this by generating user-friendly, interactive reports that summarize local and global explanations, making it easier to communicate findings. This enhanced visualization and interactivity allows for better insights into individual predictions and model behavior [36].

ELI5: ELI5 is a Python library that simplifies the interpretation of ML models by offering human-readable explanations. It supports a variety of models and algorithms, providing insights into feature importance and decision-making processes. Unlike SHAP, which is based on game theory, ELI5 can be applied to a wide range of models with minimal overhead, making it a flexible tool for quick and understandable interpretations. Its key strength is offering explanations in a way that is easily digestible by non-technical audiences.

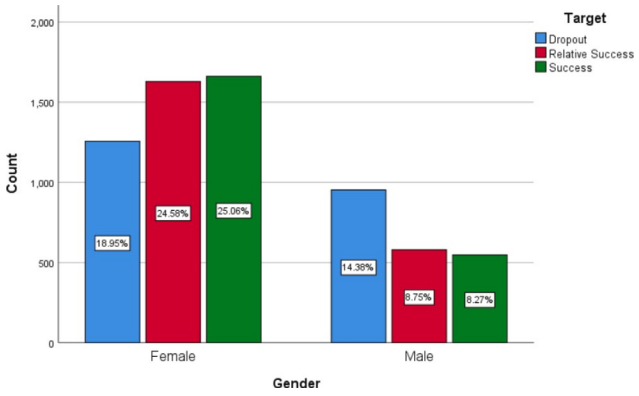


Fig. 2. Students' academic performance association with their gender.

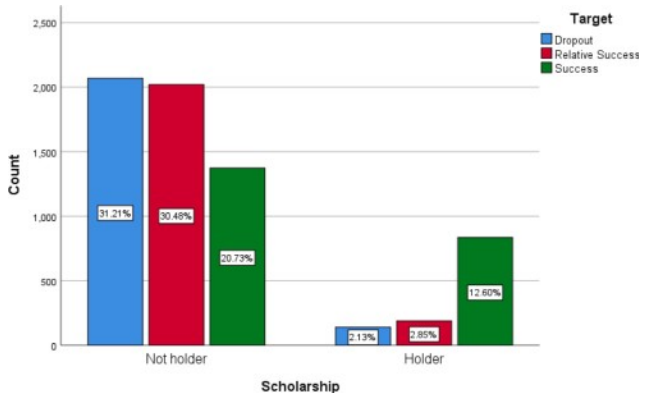


Fig. 3. Students' academic performance associated with their scholarship status.

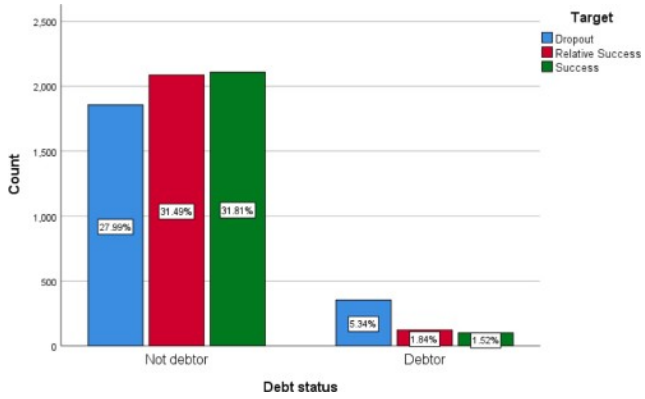


Fig. 4. Students' academic performance associated with their debt status.

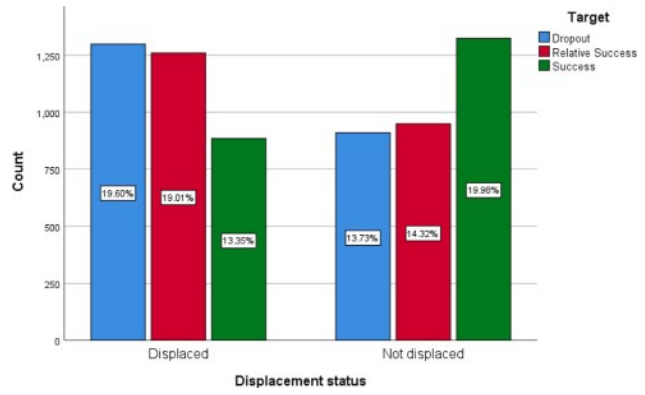


Fig. 5. Students' academic performance associated with their displacement status.

LIME: LIME (Local Interpretable Model-agnostic Explanations) explains individual predictions by approximating the black-box model locally with an interpretable surrogate model. It generates explanations specific to a single prediction by training this simple surrogate model that mimics the behavior of the complex model in a small localized area around the data point of interest. LIME is model-agnostic, meaning it can be applied to any classifier or regressor, and is particularly helpful when transparency is required for critical individual decisions, such as in healthcare or finance.

Differences of XAI Techniques: SHAP is well-suited for providing a comprehensive global understanding of a model's feature contributions. SHAPash enhances SHAP by providing interactive, user-friendly reports, making it more practical for communication with non-technical stakeholders. ELI5 is a simpler, faster approach that focuses on quick interpretability. It is useful when understanding complex models in real-time is essential without diving deep into the nuances of Shapley values. LIME is highly effective when explaining individual predictions using a surrogate model.

4. Experimental result

In this section, we describe the results obtained from the different methods applied. The bar charts plot the outcome of the chi-squared test. The charts illustrate that there is a significant variation among students' gender, scholarship status, debtor status, displaced status, tuition fee status, and academic performance. Tables 4 and 5 demonstrate the impact of balancing, preprocessing, and feature selection techniques on model performance. Figs. 8 and 9 illustrate the comparison of the proposed models with some existing models. The following figures and tables illustrate and tabulate the decision-making mechanism of the best XGB model.

4.1. Exploratory data analysis

In this section, we discuss some inner insight into our dataset the performance of the students, and some behind story.

Students' academic performance varies with their genders which can be seen in Fig. 2. Analyzing the statistics we can give some evidence to prove that earlier given statement. The chart illustrates that male student's success and relative success rates are low compared to the dropout rate. In contrast, the rate of success and relative success of the female students is high compared to the dropout rate. However, the overall percentage of females exceeds that of males. So, we can significantly differentiate students' academic performance in some aspects concerning gender.

There is a significant difference noticed in Fig. 3 between students' academic performance among those who secured their scholarships and those who could not secure scholarships. The bar chart shows that the academic success rate of scholarship holders tends to be high while the dropout rate is low. On the other hand, a different scenario has been observed for those who could not secure their scholarship. The student who did not receive scholarships had a higher dropout rate than their success rate.

As demonstrated in Fig. 4 the debt status of the students has an impact on their academic performance. Students who suffer from loan anxiety have low academic success rates and high dropout rates. In contrast, the success rate of loan-free students is high, and their dropout rate is low. It is better if they do not involve themselves in loans and debts. So, we suggest that students try to be debt-free as much as possible because the loan hurts their academic performance, which brings a downfall to their academic performance.

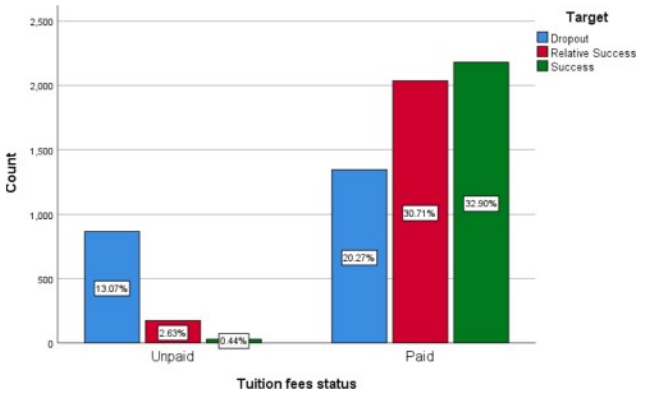


Fig. 6. Students' academic performance associated with their tuition fee status.

Table 4
Performance of the applied models before balancing, preprocessing, scaling, and feature selection.

Method	Class	Precision	Recall	F1 score	Acc.
DT	0	0.73	0.64	0.68	0.67
	1	0.38	0.40	0.37	
	2	0.76	0.79	0.78	
RF	0	0.83	0.76	0.79	0.77
	1	0.55	0.35	0.43	
	2	0.78	0.93	0.85	
GB	0	0.85	0.76	0.80	0.76
	1	0.49	0.34	0.40	
	2	0.78	0.92	0.84	
XGB	0	0.84	0.75	0.79	0.77
	1	0.52	0.43	0.47	
	2	0.79	0.91	0.85	

Fig. 5 exhibits that there is an impact of displacement on the academic performance of the students. The conclusive evidence is visible in the graph given. The success rate of displaced students is high and the dropout rate is moderate. In contrast, the dropout rate for nondisplaced students is high, but their success rate is low.

The association between the student's academic performances with their 'Tuition fees status' is visible in Fig. 6. The success rate of the students who paid their tuition fees is higher than their dropout rate. In contrast, the dropout rates for the students who did not pay their tuition fees are much higher than their success rates.

4.2. Results of the classifiers

Tables 4 and 5 tabulate the performance of the models applied in this study before and after preprocessing, feature selection, and data scaling in terms of several model performance measurement techniques such as accuracy, precision, recall, and F1 score. There is a significant impact of preprocessing, feature selection, and normalization on the model's performance that is observed.

Table 4 shows that DT is more accurate in predicting classes 0 and 2, which are Dropout and Success, compared to class 1, which is relatively successful. It also appears that the RF ensemble is more powerful than a single DT in prediction. This study obtains XGB as the best model in terms of correct predictions.

Fig. 7 summarizes the quality of the five applied models by visualizing their confusion matrix. It plots each model's accuracy and error rates in the prediction. If the predicted value is similar to the actual value that is the correct prediction. On the other hand, where the predicted and actual values do not match which means the model fails to capture the pattern for that instance. In the case of a negative actual value, if a model predicts it as positive, it is called a false positive. In contrast, if the actual value is positive and a model gives a negative prediction, that is called a false negative. The visualization exhibits that the extreme gradient boost made a smaller number of wrong predictions compared to the other three models.

Table 5
Performance of the applied models after balancing, preprocessing, scaling, and feature selection.

Method	Class	Precision	Recall	F1 score	Acc.
DT	0	0.73	0.75	0.74	0.72
	1	0.68	0.69	0.68	
	2	0.76	0.73	0.75	
RF	0	0.90	0.78	0.84	0.82
	1	0.75	0.82	0.78	
	2	0.81	0.85	0.83	
GB	0	0.86	0.74	0.79	0.79
	1	0.73	0.77	0.75	
	2	0.79	0.86	0.82	
XGB	0	0.89	0.80	0.84	0.83
	1	0.78	0.83	0.80	
	2	0.82	0.86	0.84	

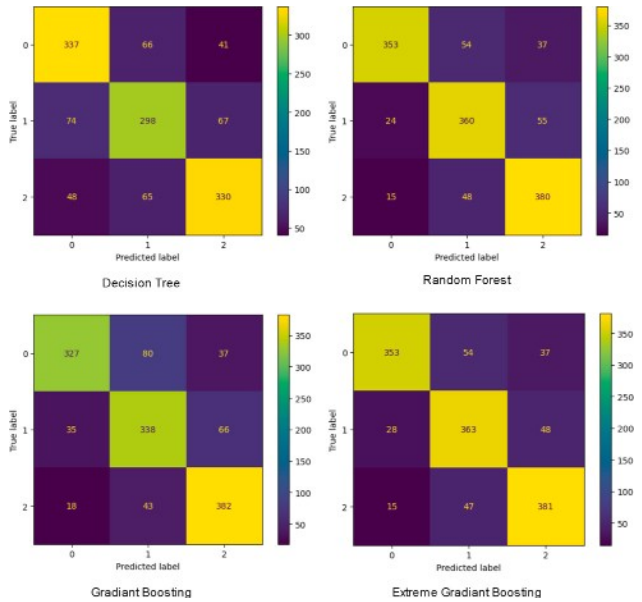


Fig. 7. Confusion matrix of the applied models.

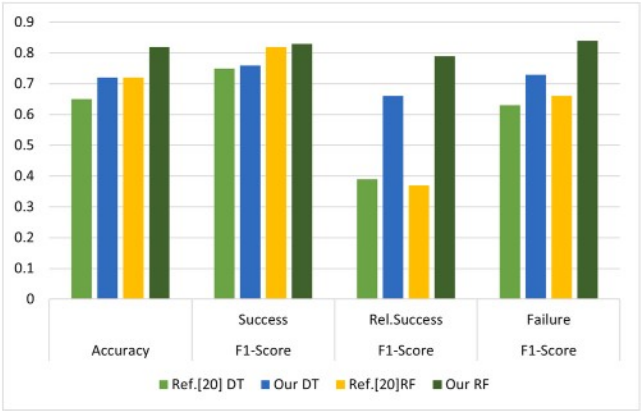


Fig. 8. Performance comparison for the standard models.

Performance comparisons of our approach with the existing baseline article approach for the two standard models are shown in Fig. 8. On the left side in the first block of the figure, it displays that our approach



Fig. 9. Performance comparison for the boosting models.

Table 6
Performance of the applied models for (50:50) ratio of training and testing of the dataset.

Method	Class	Precision	Recall	F1 score	Acc.
DT	0	0.75	0.74	0.74	0.71
	1	0.65	0.67	0.66	
	2	0.73	0.72	0.73	
RF	0	0.87	0.77	0.82	0.80
	1	0.76	0.78	0.77	
	2	0.79	0.86	0.83	
GB	0	0.85	0.74	0.79	0.78
	1	0.74	0.74	0.74	
	2	0.77	0.87	0.82	
XGB	0	0.85	0.77	0.81	0.81
	1	0.78	0.78	0.78	
	2	0.79	0.86	0.82	

achieved better accuracy than the existing one for both DT and RF models. The next three blocks of the figure depict the student's class-wise performance comparisons in terms of the F1 Score. The second left block of the figure shows that the proposed models perform slightly better than existing models in the success rate predictions of students' academic performance. The third left block of the figure shows that the proposed two models DT and RF outperform the existing two standard models. Finally, the rightmost block of the figure exhibits that in predicting failure students, our DT and RF models perform better than existing models.

Here, the performance comparisons of our two boosting models with the existing baseline article's two boosting models are illustrated in Fig. 9. On the left side in the first block of the figure appears that our XGB model achieved better accuracy than the existing model. The next three blocks of the figure display the student's class-wise performance comparisons in terms of F1-Score. The second left block of the figure shows that our model's performance is almost similar to existing models for success rate predictions of student's academic performance. The third left block of the figure shows that our XGB model outperforms the existing XGB model. Finally, the rightmost block of the figure exhibits that our models are better at predicting the failure of students accurately than existing models.

4.3. Models significant test

The results of the applied model for the 50:50 and 70:30 ratios are presented in Tables 6 and 7, respectively. Upon comparing the contents of these two tables, we can observe nearly identical results across all the methodologies employed. This consistency implies that our model performs effectively, demonstrating robust and reliable outcomes.

Table 7
Performance of the applied models for (70:30) ratio of training and testing of the dataset.

Method	Class	Precision	Recall	F1 score	Acc.
DT	0	0.76	0.74	0.75	0.71
	1	0.65	0.68	0.67	
	2	0.73	0.72	0.72	
RF	0	0.89	0.76	0.82	0.80
	1	0.75	0.81	0.78	
	2	0.78	0.85	0.82	
GB	0	0.86	0.73	0.79	0.78
	1	0.72	0.76	0.74	
	2	0.77	0.85	0.81	
XGB	0	0.87	0.78	0.82	0.82
	1	0.78	0.81	0.80	
	2	0.81	0.86	0.83	

Table 8
The 5-fold and 10-fold cross-validation results for XGB model.

XGB 5-fold		XGB 10-fold	
Fold	Accuracy	Fold	Accuracy
1	0.827	1	0.842
2	0.832	2	0.831
3	0.818	3	0.828
4	0.831	4	0.855
5	0.822	5	0.825
		6	0.822
		7	0.826
		8	0.843
		9	0.839
		10	0.846
Mean Accuracy: 0.83		Mean accuracy: 0.84	
Standard Deviation: 0.00		Standard deviation: 0.01	

4.4. Cross validation

Table 8 compares the performance of the XGB model in two cross-validation techniques: 5-fold cross-validation and 10-fold cross-validation. The accuracy is very stable throughout the folds, as indicated by the 5-fold cross-validation, with a standard deviation of 0.00. The performance variation is marginally higher, but still rather low, according to the 10-fold cross-validation standard deviation of 0.01. Both cross-validation methods demonstrate strong and stable performance, with only minor differences between individual folds.

4.5. Validation of performance differences

Our proposed models achieved better performance compared to the current models. However, the concern is whether the performance difference is statistically significant or not. we perform a paired t-test to check statistical significance. The results provide a p-value of 0.002 that demonstrates that the performance difference between the proposed models and baseline article models is statistically significant.

4.6. Explainability results

In this section, we show the result of the XAI techniques to interpret the black-box model to find the unseen story behind the prediction.

The feature importance plot in Fig. 10 shows that the number of curricular units approved by a student in the 2nd semester (last semester recorded) is the highest contribution to the student's academic

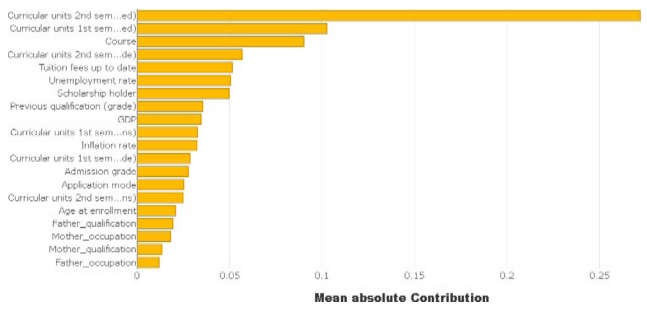


Fig. 10. Features Importance of the best model (XGB).

Table 9
Features Weight in the best model (XGB).

Feature	Weight
Curricular units 2nd sem (approved)	0.2461
Tuition fees up to date	0.1650
Scholarship holder	0.0787
Curricular units 1st sem (approved)	0.0736
Curricular units 2nd sem (evaluations)	0.0398
Course	0.0361
Curricular units 2nd sem (grade)	0.0339
Curricular units 1st sem (evaluations)	0.0332
Inflation rate	0.0313
Unemployment rate	0.0308
Age at enrollment	0.0301
Mother occupation	0.0293
GDP	0.0281
Father occupation	0.0225
Application mode	0.0216
Father qualification	0.0215
Previous qualification (grade)	0.0202
Admission grade	0.0202
Curricular units 1st sem (grade)	0.0195
Mother qualification	0.0186

performance predictors model. The plots also display that students' approved number of curricular units in the 1st semester is the 2nd highest contributor to the performance predictive model. Combinedly a student's 1st and 2nd-semester numbers of curricular units approved has a one-third contribution on the model which means more than 33% students' academic performance can be explained by observing their previous two semesters of curricular units approved. Which course a student is studying has also major influence on their academic performance. In addition, a student's previous (2nd) semester grade has some influence on their next semester's performance. Moreover, a student's tuition fees are up-to-date status, and whether a student can manage a scholarship, and their previous qualification grade has an impact on their academic performance. It is also noticed that the parents' qualifications and occupation of the students do not have much influence on their academic performance.

Table 9 demonstrates the model weights of the features to make a decision. It appears that the model provides the highest importance in the student's last semester (2nd) curricular units approved to predict student academic performance. Then the model provides the 2nd highest importance on the student's tuition fee up-to-date status.

The importance of the permutation feature is another great way to figure out the top features of the predictive models. Table 10 demonstrates the permutation feature importance of the XGB model. This technique provides the highest weight on students' number of curricular units approved in their last semester to predict their academic

Table 10
Permutation Feature Importance of the best model (XGB).

Feature	Weight
Curricular units 2nd sem (approved)	0.1970 ± 0.0121
Tuition fees up to date	0.0897 ± 0.0023
Course	0.0561 ± 0.0037
Age at enrollment	0.0323 ± 0.0026
Admission grade	0.0320 ± 0.0038
Previous qualification (grade)	0.0276 ± 0.0017
Curricular units 1st sem (approved)	0.0275 ± 0.0044
Curricular units 1st sem (grade)	0.0244 ± 0.0019
Curricular units 2nd sem (grade)	0.0220 ± 0.0013
Unemployment rate	0.0219 ± 0.0043
Curricular units 2nd sem (evaluations)	0.0196 ± 0.0032
Mother occupation	0.0183 ± 0.0025
GDP	0.0163 ± 0.0019
Application mode	0.0154 ± 0.0014
Inflation rate	0.0146 ± 0.0021
Father qualification	0.0112 ± 0.0030
Scholarship holder	0.0090 ± 0.0015
Father occupation	0.0087 ± 0.0021
Mother qualification	0.0071 ± 0.0014
Curricular units 1st sem (evaluations)	0.0069 ± 0.0016

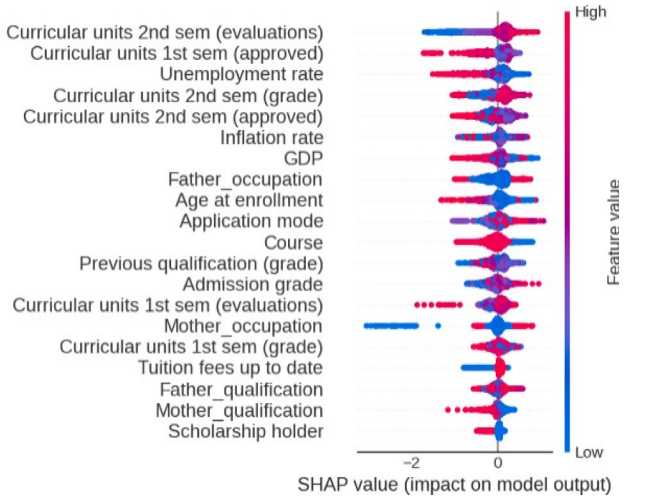


Fig. 11. Summary plot of the best model (XGB).

performance for the next semester. The post part (after plus minus) of the weights columns refers to how much performance will fluctuate if a particular feature is re-added to the model after dropping that feature from the model.

The summary plot in Fig. 11 illustrates different impacts of different features in the model. The lower value of the topmost feature 'Curricular units 2nd Sem (evaluations)' negatively impacts the model. For a higher value, it has a positive impact on the model. Meanwhile, for the top 2nd most features that show a different scenario, it hurts the model for the higher feature value and positively impacts the lower feature value. For the high value of the last feature, 'Scholarship holder' has a negative impact on the model, and for the low value, it has a positive impact on the model. Similarly, another impact of the feature on the model can be described.

Table 11
Summarization of above four XAI techniques explainability.

Features	Rank by Fig. 10	Rank by Fig. 11	Rank by Table 9	Rank by Table 10	Avg. Score	Avg. Ranking
Curricular units 2nd sem (approved)	1	5	1	1	2	1
Curricular units 1st sem (approved)	2	2	4	7	3.75	2
Course	3	11	6	3	5.75	3
Curricular units 2nd sem (grade)	4	4	7	9	6	4
Tuition fees status	5	17	2	2	6.5	5
Rate of Unemployment	6	3	10	10	7.25	6
Curricular units 2nd sem (evaluations)	15	1	5	11	8	7
Age at enrollment	16	9	11	4	10	8
Inflation rate	11	6	9	15	10.25	9
GDP	9	7	13	13	10.5	10
Previous qualification (grade)	8	12	17	6	10.75	11
Scholarship status	7	20	3	17	11.75	12
Admission grade	13	13	18	5	12.25	13
Curricular units 1st sem (evaluations)	10	14	8	20	13	14
Application mode	14	10	15	14	13.25	15
Curricular units 1st sem (grade)	12	16	19	8	13.75	16
Mother occupation	18	15	12	12	14.25	17
Father occupation	20	8	14	18	15	18
Father qualification	17	18	16	16	16.75	19
Mother qualification	19	19	20	19	19.25	20

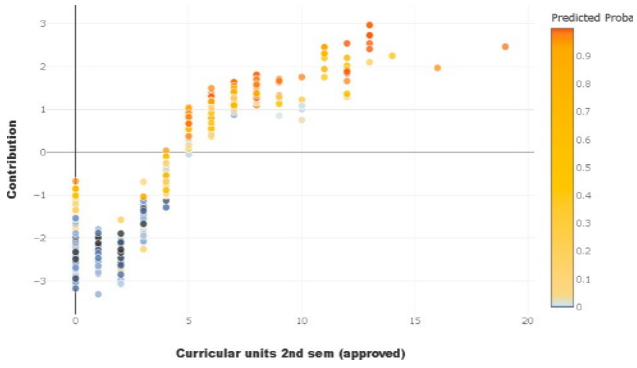


Fig. 12. Contribution plot of the students 'curricular units 2nd sem (approved)' feature.

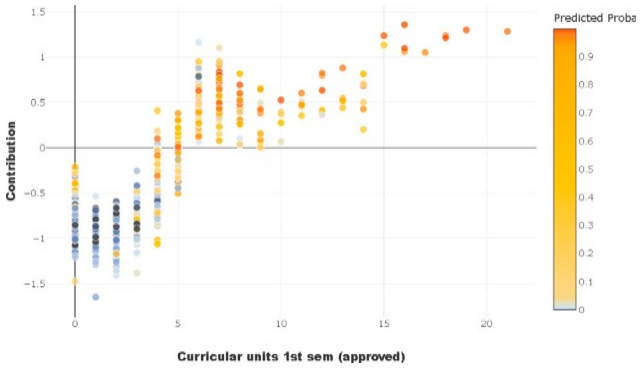


Fig. 13. Contribution plot of the students 'curricular unit's 1st sem (approved)' feature.

4.7. Generalizability of global XAI techniques

Fig. 10, Fig. 11, Table 9, and Table 10 demonstrate the overall model explainability of the best model (XGB). Table 11 summarizes the mentioned four methods and ranks general features of the underlying decision-making mechanism of the model prediction. The average ranking of the features is computed by combining the average score of the output of four applied global explainability methods. This provides a clear idea about the highly important (Avg. ranking between 1 to 5), important (6 to 10), moderate (11 to 15), and low important (16 to 20) features to predict student academic performance.

Fig. 12 shows the 'Curricular units 2nd Sem (approved)' feature contribution to the model prediction. It exhibits that for the feature value of less than 5, this feature has a negative impact on the model prediction. But for the feature value equal to or greater than 5, it has a positive impact on the model prediction.

The Curricular units 1st Semester (approved) feature contribution to the model is shown in Fig. 13. It is noticed that when a feature value is less than 5, it has a negative impact on the model except in a few cases where it does not. But in the case of feature value 5, it has both kinds of impact, in some cases positive and in other cases negative. On the other hand, a feature value greater than this feature always has a positive impact on the model.

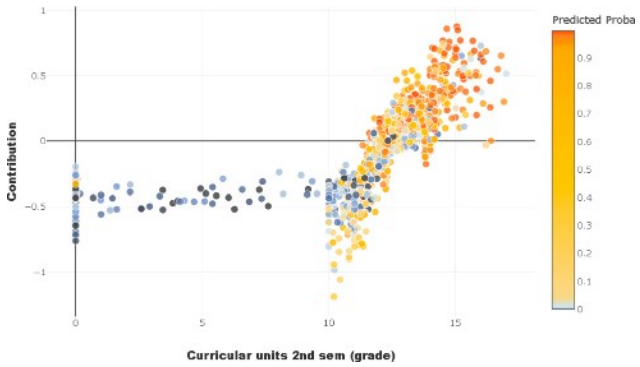


Fig. 14. Contribution plot of the students 'curricular unit's 2nd sem (grade)' feature.

The plot of the contribution of the feature Curricular unit's 2nd semester (grade) is explained in Fig. 14. The scatter plot exhibits that from the feature value of less than 10, there is a negative contribution to the model. On the other hand, when it has a feature value greater

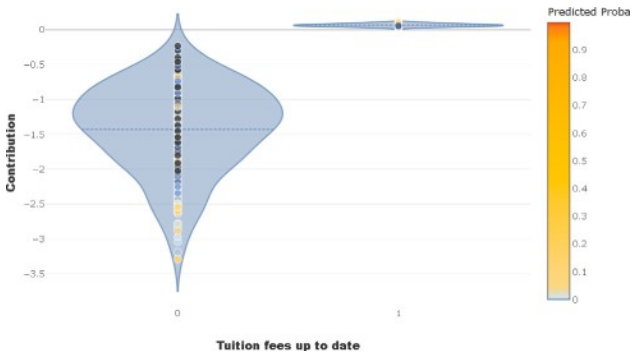


Fig. 15. Contribution plot of the students 'tuition fees' status feature.

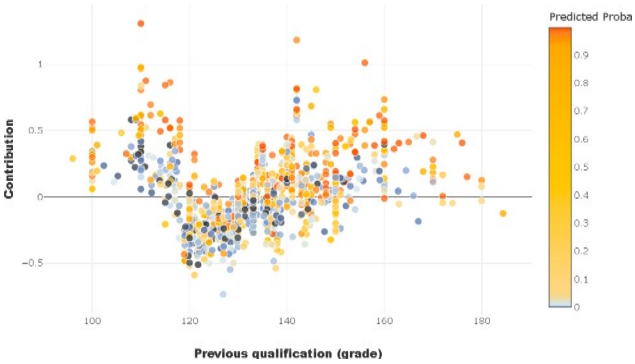


Fig. 16. Contribution plot of the students feature 'Previous qualification (grade)' feature.

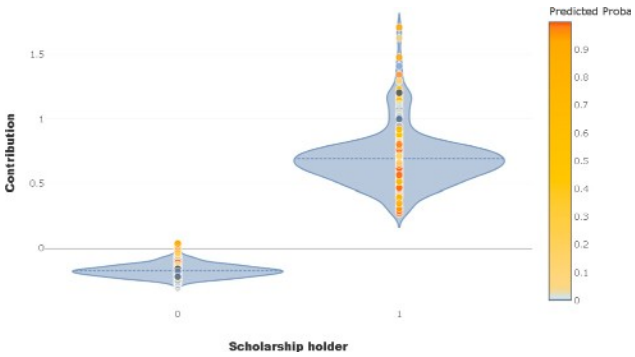


Fig. 17. Contribution plot of the student's 'scholarship status' feature.

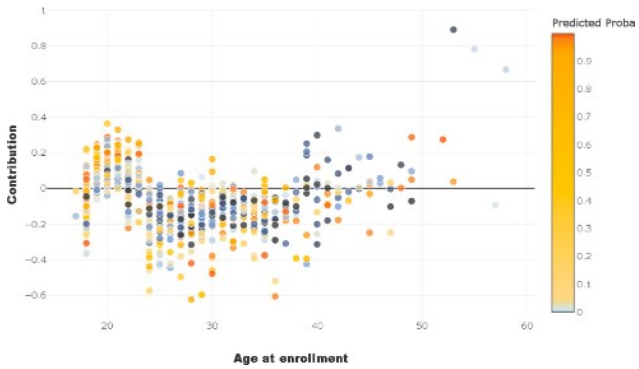


Fig. 18. Contribution plot of the students 'age at enrollment' feature.

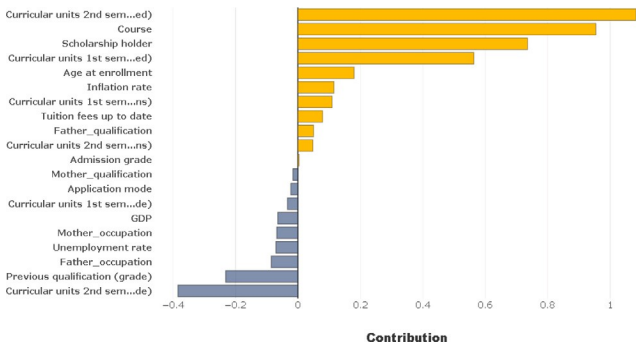


Fig. 19. A particular instance prediction explainability.

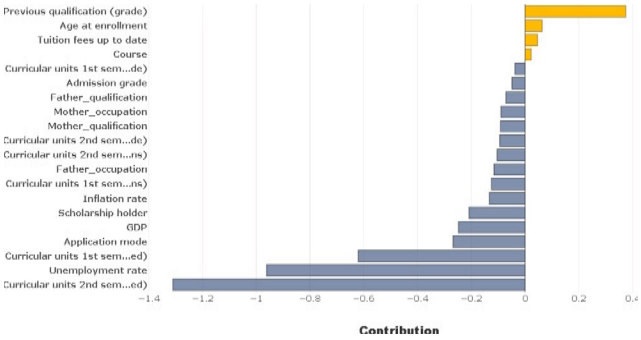


Fig. 20. A particular instance prediction explainability.

than 10, then in most cases, it has a positive impact on the model but in some cases, it also has a negative impact on the model.

The contribution of the student's 'tuition fees up to date' feature in the student's academic performance prediction model is shown in Fig. 15. It is a vital factor that has a great impact on the student's performance. For the 0 response, we can see that this feature has a negative contribution to the model. But in the case of response 1, it has a positive contribution to the model.

The student's 'Previous qualification (grade)' and its contribution to the model are depicted in Fig. 16. For grades less than 120 and greater than around 155, it has a positive contribution to the model. But in other cases, this feature has a mixed contribution, which means that in some cases it is positive and in other cases, it is negative contributions to the model.

The 'Scholarship status' feature's contribution to the model is evident in Fig. 17. For the 0 response, this feature has a negative impact on the model. But for 1 response it has a positive impact on the model.

Fig. 18 shows students' age at enrollment feature contribution to the model. It appears that young adulthood students have a positive contribution to the model which means these students' academic performance is relatively better than that of students of different ages. The scatter plot also shows for older students that academic performance is not good except for some exceptions.

Fig. 19 illustrates the explainability of an individual instance. In the given plot, the feature that is in orange color has a positive contribution to the prediction. In contrast, the features in blue color have a negative contribution to the desired prediction. The x-axis represents the amount of contribution of each feature. The larger the size of the bar chart, the greater the positive or negative influence on the prediction of the model.

Fig. 20 shows the model's explainability of an individual instance. In the plot, the feature in orange color has a positive impact on the prediction. On the other hand, the feature which is in blue colors has a negative contribution to the prediction. In general, the negative prediction dominates the ratio.



Fig. 21. Explainability of the first prediction made by the XGB model.

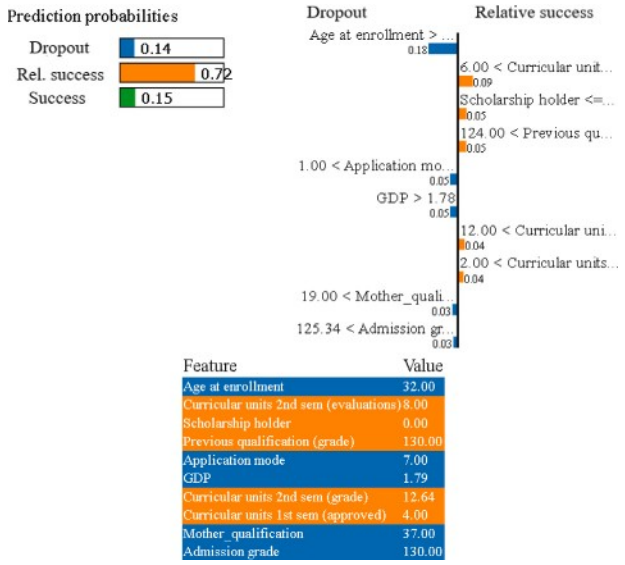


Fig. 22. Explainability of 51st prediction made by the XGB model.

Fig. 21 illustrates the explainability of an individual prediction. In the case of this prediction, the features that are in blue colors are forced to predict 0. In contrast, the features that are in orange colors are encouraging to predict 0.

Fig. 22 states the model's explanation of the ability of an individual prediction. As seen in the figure, the features in orange colors are voting to predict 1 which is the relative success class of the student's academic performance prediction model. But features in blue colors are voting out to predict 0 which is the Dropout class. The score value alongside the target categories such as Dropout, Relative Success, and Success represents the probability value of being a particular category of the considered 51st test case. However, the score value in the colors green and orange alongside the Feature column represents the value of the different features for the 51st test case prediction.

The comparison of features' impact on the model for four different instances is illustrated in Fig. 23. For id 1039 prediction all the features except the top two have positive and 0 contributions. But for the id 1795, all the features except the feature Course have a positive

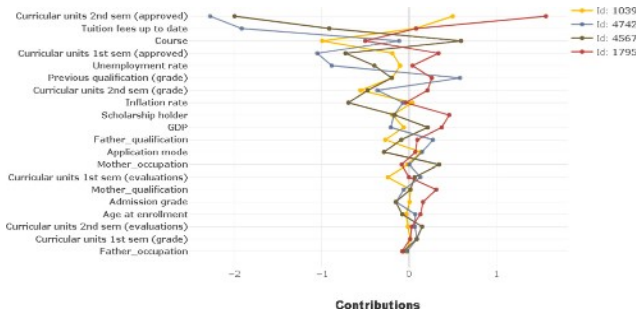


Fig. 23. Four instances model explainability comparison.

contribution to the prediction. Similarly, the prediction of the other two IDs can be described.

In summary of XAI techniques, the Table 11 presents the features ranking by the applied four global model XAI techniques and computes the average ranking by considering each technique ranking. It displays that student's last two semesters' number of total credits approved, the last semester grade and the course they are studying play a major role in the performance classification developed model. Whereas, the predictors related to the students' parents' occupation and qualifications are the bottom contributors out of 20 predictors to the student performance classification model. Other predictors have a moderate contribution to the student's performance classification model. On the other hand, an individual features different feature values contribution to the model prediction and an individual student case prediction explanation are illustrated from Figs. 13 to 23. Figs. 19 to 23 show that those features obtained as top contributors in the model prediction through global XAI techniques, which are in Figs. 10, 11, and Tables 9, 10, are also a top contributor for an individual student performance prediction.

4.8. Discussion

The results demonstrated in previous sections uncover the potential of the ML methods in EDM. The traditional way of counseling students is relatively slow and less effective. On the contrary, ML models can achieve that quickly and more effectively. The proposed model can help educators to find out weak students and decrease dropout rates. One of the most difficult tasks to achieve is to find the most appropriate model for any machine learning-based solution. This study discovered the best model by measuring the performance matrices. In general, we focus on classification using educational data in our study in comparison. Here, we evaluated both the standard and the boosting ML algorithms on a dataset from the University of Portugal. In addition, we indicated the key factors that have the greatest effect on the prediction of the results. The explainable AI (XAI) was a huge support to figure that out. As seen in previous studies [20] there are still drawbacks and a lack of effectiveness in predicting results. We have achieved better outcomes. The performance comparison plot of the models in Figs. 8 and 9 show that the proposed RF and XGB models perform 10% better than the models of the baseline article in terms of accuracy. Since the dataset contains the student's information of a single institution that is why it lacks some diversity because the student's performance influencing factors could vary with respective to different institutions. This is one of the areas that could be explored in the future and investigated to check if there are any significant variations among the factors influencing student academic performance. Consequently, the more diversified data set could ensure better generalizability of the proposed system. Though we used some of the economic factors such as GDP and inflation rate, some demographic factors such as age, gender, and parental education as the predictors while training the model as these were selected among the top 20 features out of 36. However, the XAI results in 10 and Tables

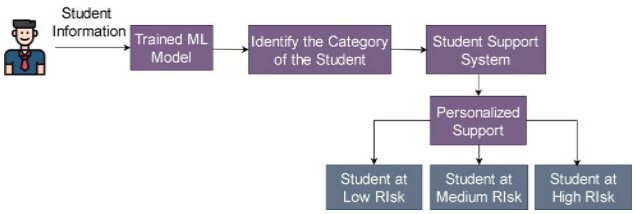


Fig. 24. Proposed student support system.

9, 10 demonstrated that these features carry very little weight and that indicates these features have no significant influence on the model prediction which may reinforce systematic inequalities in education.

5. Decision support system

The proposed support system is shown in Fig. 24. The working mechanism of the decision support system is explained in the following three steps.

5.1. Input section

Student information has been taken according to features using the Web or any Android application. It needs to be cautious while providing the information of a student. Making any mistake reflects on the prediction that leads to the wrong recommendations for that particular student, which is ultimately ineffective for that student.

5.2. Categorization section

The given information is fed to the best ML model found in this study. The model identifies the category of the student based on the experience learned during the model training phase from the academic performance data.

5.3. Personalized support section

After categorizing the students based on their predicted academic performance, personalized support is provided under the categories of students at high risk, medium risk, and low risk. This support is based on the insights derived from the explainability techniques. The XAI techniques play a crucial role in understanding the key features contributing to individual student performance. For example, in the case of high-risk students, the model identifies specific factors, such as their performance in curricular units, grades, and tuition status, that negatively impact their academic results. By pinpointing these factors, the support system can offer targeted interventions, such as academic counseling, tutoring, or financial aid assistance.

The explanations provided by XAI help tailor the support provided to each student, ensuring that the recommendations are specific to the individual's situation. For example, if a student is categorized as high risk due to poor performance in key curricular units, the system can recommend subject-specific tutoring. Similarly, if financial challenges are identified as a contributing factor, the system could suggest scholarship opportunities or financial counseling. This personalized approach ensures that both students and the institution can work together more effectively to minimize dropout rates and enhance academic performance. The combination of performance prediction and explainability allows for a more comprehensive and individualized support strategy, ensuring that the decision-making process is transparent and actionable for each student.

5.4. A practical examples of the proposed methodology in normal educational activities

In terms of practical examples, consider normal educational activities as a study area for the implementation of this methodology. We need to change some steps based on the characteristics of normal educational activities. We divide the total illustration into five major segments, namely data collection, data processing, model development and evaluation, and explainability. We have named it "Personalized Learning Path in Mathematics" Here, we have considered a few features and data samples for the practical illustration.

Step 1: Questionnaire Design and Data Collection We choose some variables for the practical example. The independent variables are 'Learning Style', 'Math Score', 'Engagement Level', 'Time Availability' and consider 'Level of Learning' as the dependent variable. Then we create synthetic data of some students for illustration of this methodology in a practical example. In Table 12, we show the variables and data items that are considered for analysis.

Step 2: Data Preprocessing After collecting the data, it needs to be preprocessed for final model training. Since variables one and four are non-numeric and categorical, we apply label encoding to convert them into numeric categorical values. Similarly, the target column is also converted into numeric categorical values to make it suitable for fitting into the ML model.

Step 3: Model Development and Evaluation After preprocessing the dataset, we separate the label column, 'Level of Learning', from the dataset and assign it to a separate variable. The dataset is then split into an 80:20 ratio for training and testing. The training dataset is fitted to the ML models, which are evaluated using performance metrics such as accuracy, precision, recall, and the F1 score to identify the best model for further analysis. This approach enables us to classify any student's level of learning effectively.

Step 4: Explainability We apply XAI tools to clearly identify why a student's learning level is categorized as low, medium, or high. These tools help pinpoint the key variables responsible for the student's current level, enabling the student to take necessary actions to maintain or improve their status. For instance, a student can dedicate more time to studying, adjust their level of engagement, modify their learning style, or aim for higher scores in math to enhance their level of learning.

Step 5: Support System From the perspective of explainability, both students and authorities can understand the reasons behind a student's current performance level. Students can receive suggestions from the support system based on their current stage using predefined tools or consult experts for more tailored advice.

The above steps clearly illustrate a practical example of the proposed methodology in regular educational activities. Domain experts or anyone interested in using this methodology can easily incorporate new variables and gain insights from these variables through our proposed method.

5.5. Challenges to apply this methodology

To apply this proposed methodology in real-life applications, several major challenges need to be addressed. The challenges in detail are discussed. The first challenge in implementing the proposed methodology is the absence of an organized dataset. Most publicly available datasets are not structured in a way that facilitates the application of the proposed methodology. In addition, these datasets often lack sufficient information to develop an effective system for any specific institution without additional explicit knowledge. Furthermore, it is challenging to collect data from participants due to their changing willingness to participate, and it has become increasingly rare for individuals to voluntarily provide data., Second, as this is a sensitive domain, the application of the proposed methodology's outcomes must be validated by experienced academicians and domain experts. Furthermore, there should be several trial versions of the methodology

Table 12
Variables name and sample values.

Variable Name	Description	Sample Values
Learning Style	Preferred learning style of the Student	'Visual', 'Auditory', 'Kinesthetic'
Math Score	Math scores achieved by the student (out of 100)	50, 60, 85, 100
Engagement Level	Engagement level with learning activities (scale: 1-10)	1, 5, 7, 10
Time Availability	Time available for learning each day	'<1 hour', '1-2 hours', '>2 hours'
Level of Learning	Overall learning level (Low, Medium, High)	'Low', 'Medium', 'High'

conducted before its practical implementation with students. Finally, the process requires expert personnel for requirement analysis, data collection, data analysis, and policy development. These tasks are both time-consuming and costly. Without external funding, it will be very challenging to implement the methodology effectively.

6. Conclusion and future works

In this study, we started with data preprocessing and then performed EDA on the set of variables. We observed how different attributes such as gender, scholarship status, debt status, displaced, and tuition fee up-to-date status are significantly influence the academic performance of students and interpreted the insight. Then, we performed some data preprocessing to make the data suitable for building ML models measuring the performance of the applied models, and comparing them with the existing works. Eventually, we obtained the result that both the standard model and the boosting model achieved better performance than the existing models. In the end, we used various XAI tools to reveal the decision-making mechanism of the applied models in the back. Finally, a student support system is proposed that utilizes the dataset, divides academic performance into categories, and ensures specific support to students.

A future direction for this research is the qualitative assessment of XAI methods outcome by experienced educators. Additionally, the integration of diverse data collection methods beyond relying solely on direct responses from students. For instance, incorporating real-time data gathered through computer vision systems or other advanced approaches, such as monitoring students during specific activities, could significantly enrich the dataset. By combining traditional self-reported data with additional behavioral and contextual characteristics, the ability of the model to classify and predict students' academic performance can be substantially enhanced. This multifaceted approach would not only improve the accuracy and robustness of the model but would also enable authorities to offer more personalized and effective support tailored to each student's unique needs and challenges.

CRediT authorship contribution statement

Md. Mahmudul Islam: Writing – original draft, Visualization, Validation, Methodology, Formal analysis. **Farhad Hossain Sojib:** Writing – original draft, Formal analysis, Data curation. **Md. Fazle Hasan Mihad:** Writing – original draft, Visualization, Methodology, Formal analysis. **Mahmudul Hasan:** Writing – review & editing, Supervision, Conceptualization. **Mahfujur Rahman:** Writing – review & editing, Supervision, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset used in this study is publicly available at https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success.

References

[1] Engr Sana Bhutto, Isma Farah Siddiqui, Qasim Ali Arain, Maleeha Anwar, Predicting students' academic performance through supervised machine learning, in: 2020 International Conference on Information Science and Communication Technology, ICISCT, IEEE, 2020, pp. 1–6.
[2] Maria Tsiakmaki, Georgios Kostopoulos, Sotiris Kotsiantis, Omiros Ragos, Implementing AutoML in educational data mining for prediction tasks, Appl. Sci. 10 (1) (2019) 90.
[3] Carlos Márquez-Vera, Alberto Cano, Cristobal Romero, Amin Yousef Mohammad Noaman, Habib Mousa Fardoun, Sebastian Ventura, Early dropout prediction using data mining: a case study with high school students, Expert Syst. 33 (1) (2016) 107–124.
[4] Julie L. Harvey, Sathish A.P. Kumar, A practical model for educators to predict student performance in K-12 education using machine learning, in: 2019 IEEE Symposium Series on Computational Intelligence, SSCI, IEEE, 2019, pp. 3004–3011.
[5] Ying Zhang, Samia Oussena, Tony Clark, Hyeonsook Kim, Use data mining to improve student retention in higher education-A case study, in: ICEIS (1), 2010, pp. 190–197.
[6] Manjari Chitti, Padmini Chitti, Manoj Jayabalan, Need for interpretable student performance prediction, in: 2020 13th International Conference on Developments in ESystems Engineering, DeSE, IEEE, 2020, pp. 269–272.
[7] Yazan A Alsariera, Yahia Baashar, Gamal Alkawsi, Abdulsalam Mustafa, Ammar Ahmed Alkahtani, Nor'ashikin Ali, et al., Assessment and evaluation of different machine learning algorithms for predicting student performance, Comput. Intell. Neurosci. 2022 (2022).
[8] Sarfraz Nawaz Brohi, Thulasyammal Ramiah Pillai, Sukhminder Kaur, Harsimren Kaur, Sanath Sukumaran, David Asirvatham, Accuracy comparison of machine learning algorithms for predictive analytics in higher education, in: Emerging Technologies in Computing: Second International Conference, ICETiC 2019, London, UK, August 19–20, 2019, Proceedings 2, Springer, 2019, pp. 254–261.
[9] Anal Acharya, Devadatta Sinha, Early prediction of students performance using machine learning techniques, Int. J. Comput. Appl. 107 (1) (2014) 37–43.
[10] Nabila Sghir, Amina Adadi, Mohammed Lahmer, Recent advances in predictive learning analytics: A decade systematic review (2012–2022), Educ. Inf. Technol. 28 (7) (2023) 8299–8333.
[11] Yahia Baashar, Gamal Alkawsi, Nor'ashikin Ali, Hitham Alhussian, Hussein T. Bahbouh, Predicting student's performance using machine learning methods: A systematic literature review, in: 2021 International Conference on Computer & Information Sciences, ICCOINS, IEEE, 2021, pp. 357–362.
[12] Francis Ofori, Elizaphan Maina, Rhoda Gitonga, Using machine learning algorithms to predict students' performance and improve learning outcome: A literature based review, J. Inf. Technol. 4 (1) (2020) 33–55.
[13] Mojisola G. Asogbon, Oluwarotimi W. Samuel, Mumini O. Omisore, Bolanle A. Ojokoh, A multi-class support vector machine approach for students academic performance prediction, Int. J. Multidiscip. Curr. Res. 4 (2016) 210–215.
[14] Hussein Altabrawee, Osama Abdul Jaleel Ali, Samir Qaisar Ajmi, Predicting students' performance using machine learning techniques, J. Univ. Babylon Pure Appl. Sci. 27 (1) (2019) 194–205.
[15] Leena H. Alamri, Ranim S. Almuslim, Mona S. Alotibi, Dana K. Alkadi, Irfan Ullah Khan, Nida Aslam, Predicting student academic performance using support vector machine and random forest, in: Proceedings of the 2020 3rd International Conference on Education Technology Management, 2020, pp. 100–107.
[16] Alberto Rivas, Jesús M Fraile, Pablo Chamoso, Alfonso González-Briones, Sara Rodríguez, Juan M. Corchado, Students performance analysis based on machine learning techniques, in: Learning Technology for Education Challenges: 8th International Workshop, LTEC 2019, Zamora, Spain, July 15–18, 2019, Proceedings 8, Springer, 2019, pp. 428–438.

[17] Phauk Sokkhey, Takeo Okazaki, Hybrid machine learning algorithms for predicting academic performance, Int. J. Adv. Comput. Sci. Appl. 11 (1) (2020) 32–41.
[18] Iti Burman, Subhranil Som, Predicting students academic performance using support vector machine, in: 2019 Amity International Conference on Artificial Intelligence, AICAI, IEEE, 2019, pp. 756–759.
[19] Ali Salah Hashim, Wid Akeel Awadh, Alaa Khalaf Hamoud, Student performance prediction model based on supervised machine learning algorithms, in: IOP Conference Series: Materials Science and Engineering, vol. 928, no. 3, IOP Publishing, 2020, 032019.
[20] Mónica V. Martins, Daniel Tolledo, Jorge Machado, Luís M.T. Baptista, Valentim Realinho, Early prediction of student's performance in higher education: A case study, in: Trends and Applications in Information Systems and Technologies: Volume 1 9, Springer, 2021, pp. 166–175.
[21] Mustafa Yağcı, Educational data mining: prediction of students' academic performance using machine learning algorithms, Smart Learn. Environ. 9 (1) (2022) 11.
[22] Pratiyush Guleria, Manu Sood, Explainable AI and machine learning: performance evaluation and explainability of classifiers on educational data mining inspired career counseling, Educ. Inf. Technol. 28 (1) (2023) 1081–1116.
[23] Prasadini Padmasiri, Sanvitha Kasthuriarachchi, Interpretable prediction of student dropout using explainable AI models, in: 2024 International Research Conference on Smart Computing and Systems Engineering, Vol. 7, SCSE, IEEE, 2024, pp. 1–7.
[24] Marcell Nagy, Roland Molontay, Interpretable dropout prediction: towards XAI-based personalized intervention, Int. J. Artif. Intell. Educ. 34 (2) (2024) 274–300.
[25] Radhika R. Halde, Application of machine learning algorithms for betterment in education system, in: 2016 International Conference on Automatic Control and Dynamic Optimization Techniques, ICACDOT, IEEE, 2016, pp. 1110–1114.
[26] H.P. Vinutha, B. Poornima, B.M. Sagar, Detection of outliers using interquartile range technique from intrusion dataset, in: Information and Decision Sciences: Proceedings of the 6th International Conference on Ficta, Springer, 2018, pp. 511–518.
[27] Mahmudul Hasan, Md Fazle Rabbi, Md Nahid Sultan, Adiba Mahjabin Nitu, Md Palash Uddin, A novel data balancing technique via resampling majority and minority classes toward effective classification, TELKOMNIKA (Telecommun. Comput. Electron. Control) 21 (6) (2023) 1308–1316.
[28] Nahid Sultan, Mahmudul Hasan, Md Ferdous Wahid, Hasi Saha, Ahsan Habib, Cesarean section classification using machine learning with feature selection, data balancing and explainability, IEEE Access (2023).
[29] L. Yogesh, M. Arunadevi, C.P.S. Prakash, Predicton of MRR & surface roughness in wire EDM machining using decision tree and naive Bayes algorithm, in: 2021 International Conference on Emerging Smart Computing and Informatics, ESCI, IEEE, 2021, pp. 527–532.
[30] Mahmudul Hasan, Ashraful Islam, Ashrafuzzaman Sohag, On the selection of suitable dimensionality reduction and data balancing techniques to classify DarkNet access on CICDarknet2020, in: Cyber Security and Business Intelligence, Routledge, 2023, pp. 154–173.
[31] Utkarsh Mahadeo Khaire, R. Dhanalakshmi, Effects of random forest parameters in the selection of biomarkers, Comput. J. 64 (12) (2021) 1840–1847.
[32] Nagaraj Honnikoll, Ishwar Baidari, Mean error rate weighted online boosting method, Comput. J. 66 (1) (2023) 1–15.
[33] Md Mahedi Hassan, Md Fahim Abrar, Mahmudul Hasan, An explainable AI-driven machine learning framework for cybersecurity anomaly detection, in: Cyber Security and Business Intelligence, Routledge, 2023, pp. 197–219.
[34] Ravindra Kumar Singh, Harsh Kumar Verma, Redis-based messaging queue and cache-enabled parallel processing social media analytics framework, Comput. J. 65 (4) (2022) 843–857.
[35] Mahmudul Hasan, Priyanka Roy, Adiba Mahjabin Nitu, Cervical cancer classification using machine learning with feature importance and model explainability, in: 2022 4th International Conference on Electrical, Computer & Telecommunication Engineering, ICECTE, IEEE, 2022, pp. 1–4.
[36] Sad Wadi Sajid, KM Rashid Anjum, Md Al-Shaharia, Mahmudul Hasan, Investigating machine learning algorithms with model explainability for network intrusion detection, in: Cyber Security and Business Intelligence, Routledge, 2023, pp. 121–136.